Predict the output of each task using pen and paper work then cross check it with system result. Need to identify the preprocessor stage result also.

Note: Don't execute the task directly in system.

1.

```
#define CUBE(x) x*x*x
int main()
{
int a,b=3;
a=CUBE(b++);
printf("%d %d",a,b);
}
```

2.

```
#include<stdio.h>
#define x 5+5
int main()
{
int i;
i = x * x * x;
printf("%d ",i);
}
```

3.

```
#include<stdio.h>
#define _vec(x) ++x * ++x
int main()
{
int a = 3, z;
z = ++a * ++a;
a -= 3;
printf("%d %d", _vec(a), z);
}
```

4.

```
#include<stdio.h>
#define loop while(1)
int main()
{
loop
printf("vector india ");
}
```

5.

```
#include<stdio.h>
#define int double
main()
{
int confident=12345;
printf("%d ",sizeof(confident));
}
```

6.

```
#include<stdio.h>
#define NODE(a) a*10+2
void main()
{
int a=NODE(2)*2;
printf("%d ",a);
}
```

7.

```
#include<stdio.h>
#define LOOP(a) int i;\
for(i=1;i<=a;i++);\
printf("%d ",i);
void main(){
LOOP(5);
}
```

8.

```
#include<stdio.h>
#define v 9
main()
{
int b;
#define v 10
b=v*5;
printf("%d",b);
}
```

9.

```
#define big(a,b) a > b ? a : b
#define swap(a,b) temp=a; a=b; b=temp;
void main()
{
int a=3,b=5,temp=0;
if ((3+big(a,b)) > b) swap(a,b);
printf("%d %d",a,b);
}
```

10.

```
#define calc(a, b) (a * b) / (a - b)
void main()
{
int a = 20, b = 10;
printf("%d", calc(a + 4, b -2));
}
```

11.

```
#include<stdio.h>
#define x 10;
int main(){
printf("%d",x);
}
```

12.

```
#define a 10
int main()
{
  #ifdef a
        printf("VECTOR ");
  #endif
  #ifndef a
        printf("INDIA\n");
  #else
        printf("HYDERABAD\n");
  #endif
  return 0;
}
```

13.

```
#include<stdio.h>
#define max 100
int main()
{
        #ifdef max
                printf("Hello");
        return 0;
}
```

14.

```
#include<stdio.h>
#define max 100
int main()
{
        #if(max%10)
                printf("VECTOR ");
        #endif
        printf("INDIA");
```

```
    return 0;
}
```

15.

```
#include<stdio.h>
#if X == 3
   #define Y 3
#else
   #define Y 5
#endif
int main()
{
   printf("%d", Y);
   return 0;
}
```

16.

```
#include<stdio.h>
#define X 3
#if !X
   printf("Good");
#else
   printf("Bad");
#endif
int main()
{
    return 0;
}
```

17.

```
#include <stdio.h>
#define ISEQUAL(A, B) A == B
#define VAL !1
int main()
{
```

```
    #if ISEQUAL(VAL, 0)
        printf("OK");
    #else
        printf("NOT OK");
    #endif
    return 0;
}
```

18.

```
#include <stdio.h>
#define MAX 1000
int main()
{
   int MAX = 100;
   printf("%d ", MAX);
   return 0;
}
```

19.

```
#include<stdio.h>
#define A -B
#define B -C
#define C 5

int main()
{
  printf("The value of A is %d\n", A);
  return 0;
}
```

20.

```
#include<stdio.h>
#define hello(x) printf("a"#x" = %d",a##x)
#define A3
int a3=45;
```

```
int main()
{
	#ifdef A3
		hello(3);
	#else
		printf("VECTOR");
	#endif
	return 0;
}
```

21.

```
#include<stdio.h>
#define hello(x) printf("a"#x" = %d",a##x)
int a3=45;

int main()
{
	#ifdef a3
		hello(3);
	#else
		printf("VECTOR");
	#endif

	return 0;
}
```

22.

```
#define hello(v) #v
int main()
{
	printf(hello(VECTOR, HYDERABAD));
	retirn 0;
}
```

23.

```
#define print(x,y) #x #y
int main()
{
        printf(print(hello,"World!"));
        return 0;
}
```

24.

```
#define VAL 10
int main()
{
#ifdef VAL
#undef VAL
#define VAL 100
#else
#define VAL 2000
#endif
        printf("%d\n",VAL);
}
```

25.

```
void fun()
{
#define VAL 100
        printf("%d\n",VAL);
}
int main()
{
#define VAL 200;
        fun();
        printf("%d\n",VAL);
}
```

26.

```
#define print(x) printf(#x "...")
int main()
{
        print(Hi);
        print(Good Morning!);
}
```

27.

```
#define p(n,m) printf("%d",m##n)
#define q(a,b) printf("%d",a##b)
main()
{
  p(3,4);
  q(5,6);
}
```

28.

```
#define hello 10
void main()
{
   printf("%d",hello);
   #undef hello
   printf("%d",hello);
}
```

29.

```
#define sf 10
main()
{
   if(sf==100)
      printf("good");
   else
   {
      printf("bad");
```

```
        sf=100;
    }
    printf("%d",sf);
}
```

30.

```
# define a 10
void main()
{
    printf("%d..", a);
    foo();
    printf("%d", a);
}
void foo()
{
   #undef a
   #define a 50
}
```